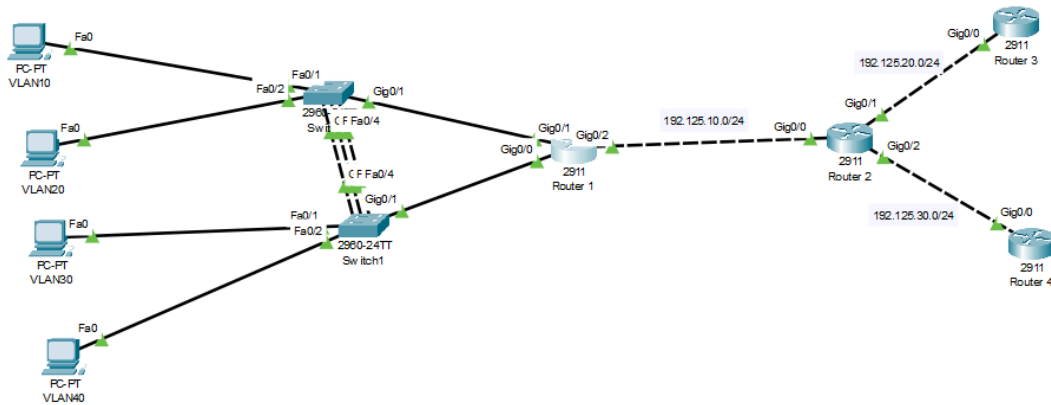


READ ME

TOPOLOGY



IP ADDRESSES

VLAN 10 → 10.10.10.0/24 (PC1)

VLAN 20 → 10.10.20.0/24 (PC2)

VLAN 30 → 10.20.30.0/24 (PC3)

VLAN 40 → 10.20.40.0/24 (PC4)

WAN R1–R2 → 192.125.10.0/30

R2–R3 → 192.125.20.0/30

R2–R4 → 192.125.30.0/30

DEVICES USED

4 ROUTERS

2 SWITCHES

4PCs

CONFIGURATIONS

- EIGRP
- STP
- EtherChannel
- HSRP
- ACL
- NAT
- BGP

ROUTERS

ROUTER 1

hostname Router

```
ip dhcp pool vlan10
network 10.10.10.0 255.255.255.0
ip dhcp pool vlan20
ip dhcp pool VLAN10
network 10.10.10.0 255.255.255.0
default-router 10.10.10.1
ip dhcp pool VLAN20
network 10.10.20.0 255.255.255.0
default-router 10.10.20.1
ip dhcp pool VLAN30
network 10.10.30.0 255.255.255.0
default-router 10.10.30.1
ip dhcp pool VLAN40
network 10.10.40.0 255.255.255.0
default-router 10.10.40.1
```

```
interface GigabitEthernet0/0
no ip address
ip nat inside
duplex auto
speed auto
```

```
interface GigabitEthernet0/0.10
no ip address
ip access-group 110 in
```

```
interface GigabitEthernet0/0.30
encapsulation dot1Q 30
ip address 10.10.30.1 255.255.255.0
```

```
interface GigabitEthernet0/0.40
encapsulation dot1Q 40
ip address 10.10.40.1 255.255.255.0
```

```
interface GigabitEthernet0/1
no ip address
ip nat inside
```

```
interface GigabitEthernet0/1.10
encapsulation dot1Q 10
ip address 10.10.10.1 255.255.255.0
ip access-group 110 in
standby 1 ip 10.10.10.254
standby 1 priority 110
```

```
standby 1 preempt
```

```
interface GigabitEthernet0/1.20  
encapsulation dot1Q 20  
ip address 10.10.20.1 255.255.255.0
```

```
interface GigabitEthernet0/2  
ip address 192.125.10.1 255.255.255.0  
ip nat outside
```

```
router eigrp 100  
network 10.0.0.0  
network 192.125.10.0
```

```
router bgp 65001  
neighbor 192.125.10.2 remote-as 65002  
network 10.0.0.0
```

```
ip nat inside source list 1 interface GigabitEthernet0/1 overload  
ip nat inside source static 10.10.40.2 200.1.1.10  
ip classless
```

```
access-list 110 deny tcp 10.10.10.0 0.0.0.255 10.10.40.0 0.0.0.255 eq www  
access-list 110 permit ip any any  
access-list 1 permit 10.0.0.0 0.255.255.255
```

ROUTER 2

```
hostname Router
```

```
interface GigabitEthernet0/0  
ip address 192.125.10.2 255.255.255.0
```

```
interface GigabitEthernet0/0.10  
encapsulation dot1Q 10  
ip address 10.10.10.2 255.255.255.0  
standby 1 ip 10.10.10.254
```

```
interface GigabitEthernet0/1  
ip address 192.125.20.1 255.255.255.0
```

```
interface GigabitEthernet0/2  
ip address 192.125.30.1 255.255.255.0
```

```
router eigrp 100  
network 192.125.10.0  
network 192.125.20.0  
network 192.125.30.0
```

```
router bgp 65002
neighbor 192.125.10.1 remote-as 65001
```

ROUTER 3

```
hostname Router
```

```
interface GigabitEthernet0/0
ip address 192.125.20.2 255.255.255.0
```

```
interface GigabitEthernet0/1
```

```
interface GigabitEthernet0/2
```

```
interface Vlan1
```

```
router eigrp 100
network 192.125.20.0
```

ROUTER 4

```
hostname Router
```

```
interface GigabitEthernet0/0
ip address 192.125.30.2 255.255.255.0
```

```
interface GigabitEthernet0/1
```

```
interface GigabitEthernet0/2
```

```
router eigrp 100
network 192.125.30.0
```

SWITCHES

SWITCH_0

hostname Switch

interface Port-channel1
switchport mode trunk

interface FastEthernet0/1
switchport access vlan 10
switchport mode access

interface FastEthernet0/2
switchport access vlan 20
switchport mode access

interface FastEthernet0/3
switchport mode trunk
channel-group 1 mode active

interface FastEthernet0/4
switchport mode trunk
channel-group 1 mode active

interface GigabitEthernet0/1
switchport trunk allowed vlan 10,20
switchport mode trunk

SWITCH_1

hostname Switch

interface Port-channel1
switchport mode trunk

interface FastEthernet0/1
switchport access vlan 30
switchport mode access

interface FastEthernet0/2
switchport access vlan 40
switchport mode access

interface FastEthernet0/3
switchport mode trunk
channel-group 1 mode active

interface FastEthernet0/4
switchport mode trunk


```
channel-group 1 mode active
```

```
interface GigabitEthernet0/1  
switchport trunk allowed vlan 30,40  
switchport mode trunk
```

TESTS

ROUTER_1

Do show ip eigrp neighbor

```
Router(config)#DO show ip eigrp neighbors
IP-EIGRP neighbors for process 100
H   Address          Interface    Hold Uptime    SRTT    RTO    Q    Seq
      (sec)              (ms)
0   192.125.10.2      Gig0/2       10   06:07:17    40     1000   0    69
```

Do sh ip route

```
Router(config)#DO show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
C       10.10.10.0/24 is directly connected, GigabitEthernet0/1.10
L       10.10.10.1/32 is directly connected, GigabitEthernet0/1.10
C       10.10.20.0/24 is directly connected, GigabitEthernet0/1.20
L       10.10.20.1/32 is directly connected, GigabitEthernet0/1.20
C       10.10.30.0/24 is directly connected, GigabitEthernet0/0.30
L       10.10.30.1/32 is directly connected, GigabitEthernet0/0.30
C       10.10.40.0/24 is directly connected, GigabitEthernet0/0.40
L       10.10.40.1/32 is directly connected, GigabitEthernet0/0.40
    192.125.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.125.10.0/24 is directly connected, GigabitEthernet0/2
L       192.125.10.1/32 is directly connected, GigabitEthernet0/2
D       192.125.20.0/24 [90/3072] via 192.125.10.2, 06:18:38, GigabitEthernet0/2
D       192.125.30.0/24 [90/3072] via 192.125.10.2, 06:18:38, GigabitEthernet0/2
```

Do sh ip bgp summary

```
Router(config)#do sh ip bgp summary
BGP router identifier 192.125.10.1, local AS number 65001
BGP table version is 2, main routing table version 6
0 network entries using 0 bytes of memory
0 path entries using 0 bytes of memory
0/0 BGP path/bestpath attribute entries using 0 bytes of memory
0 BGP AS-PATH entries using 0 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 32 total bytes of memory
BGP activity 0/0 prefixes, 0/0 paths, scan interval 60 secs

Neighbor      V    AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
192.125.10.2   4 65002    475     480        2    0    0 06:36:47      4
```

Do sh ip nat translations

```
Router(config)#do sh ip nat translations
Pro  Inside global      Inside local      Outside local      Outside global
---  200.1.1.10           10.10.40.2        ---                ---
```

Do sh access-lists

```
Router(config)#do sh access-lists
Extended IP access list 110
  10 deny tcp 10.10.10.0 0.0.0.255 10.10.40.0 0.0.0.255 eq www
  20 permit ip any any (2 match(es))
Standard IP access list 1
  10 permit 10.0.0.0 0.255.255.255
```

ROUTER_2

Do show ip eigrp neighbor

```
Router(config)#DO SH IP EIGRP NEIGHBOR
IP-EIGRP neighbors for process 100
H   Address           Interface           Hold Uptime        SRTT   RTO   Q   Seq
                               (sec)              (ms)                Cnt   Num
0   192.125.30.2       Gig0/2              14   06:10:38   40    1000   0   60
1   192.125.10.1       Gig0/0              12   06:10:38   40    1000   0   26
2   192.125.20.2       Gig0/1              10   06:10:38   40    1000   0   59
```

Do sh ip route

```
Router(config)#do sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

Gateway of last resort is not set

```
10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C    10.10.10.0/24 is directly connected, GigabitEthernet0/0.10
L    10.10.10.2/32 is directly connected, GigabitEthernet0/0.10
D    10.10.20.0/24 [90/28416] via 192.125.10.1, 06:20:43, GigabitEthernet0/0
D    10.10.30.0/24 [90/28416] via 192.125.10.1, 06:20:43, GigabitEthernet0/0
D    10.10.40.0/24 [90/28416] via 192.125.10.1, 06:20:43, GigabitEthernet0/0
192.125.10.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.125.10.0/24 is directly connected, GigabitEthernet0/0
L    192.125.10.2/32 is directly connected, GigabitEthernet0/0
192.125.20.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.125.20.0/24 is directly connected, GigabitEthernet0/1
L    192.125.20.1/32 is directly connected, GigabitEthernet0/1
192.125.30.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.125.30.0/24 is directly connected, GigabitEthernet0/2
L    192.125.30.1/32 is directly connected, GigabitEthernet0/2
```

Do sh ip bgp summary

```
Router(config)#do sh ip bgp summary
BGP router identifier 192.125.30.1, local AS number 65002
BGP table version is 1, main routing table version 6
0 network entries using 0 bytes of memory
0 path entries using 0 bytes of memory
0/0 BGP path/bestpath attribute entries using 0 bytes of memory
0 BGP AS-PATH entries using 0 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 32 total bytes of memory
BGP activity 0/0 prefixes, 0/0 paths, scan interval 60 secs

Neighbor      V    AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
192.125.10.1   4 65001   474     483       1    0    0 06:38:22      4
```

ROUTER_3

Do show ip eigrp neighbor

```
Router(config)#DO SHOW IP EIGRP NEIGHBOR
IP-EIGRP neighbors for process 100
H   Address          Interface      Hold Uptime      SRTT   RTO   Q   Seq
                               (sec)          (ms)    Cnt  Num
0   192.125.20.1      Gig0/0         13    06:12:42    40     1000   0   68
```

Do sh ip route

```
Router(config)#do sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/24 is subnetted, 4 subnets
D    10.10.10.0/24 [90/28672] via 192.125.20.1, 06:22:05, GigabitEthernet0/0
D    10.10.20.0/24 [90/28672] via 192.125.20.1, 06:22:05, GigabitEthernet0/0
D    10.10.30.0/24 [90/28672] via 192.125.20.1, 06:22:05, GigabitEthernet0/0
D    10.10.40.0/24 [90/28672] via 192.125.20.1, 06:22:05, GigabitEthernet0/0
D    192.125.10.0/24 [90/3072] via 192.125.20.1, 06:22:05, GigabitEthernet0/0
192.125.20.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.125.20.0/24 is directly connected, GigabitEthernet0/0
L    192.125.20.2/32 is directly connected, GigabitEthernet0/0
D    192.125.30.0/24 [90/3072] via 192.125.20.1, 06:22:05, GigabitEthernet0/0
```

ROUTER_4

Do show ip eigrp neighbor

```
Router(config)#DO SH IP EIGRP NEIGHBOR
IP-EIGRP neighbors for process 100
H   Address           Interface      Hold Uptime    SRTT    RTO    Q    Seq
   (sec)              (ms)          Cnt  Num
0   192.125.30.1       Gig0/0         14   06:13:38    40     1000   0    68
```

Do sh ip route

```
Router(config)#do sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/24 is subnetted, 4 subnets
D       10.10.10.0/24 [90/28672] via 192.125.30.1, 06:22:50, GigabitEthernet0/0
D       10.10.20.0/24 [90/28672] via 192.125.30.1, 06:22:50, GigabitEthernet0/0
D       10.10.30.0/24 [90/28672] via 192.125.30.1, 06:22:50, GigabitEthernet0/0
D       10.10.40.0/24 [90/28672] via 192.125.30.1, 06:22:50, GigabitEthernet0/0
D       192.125.10.0/24 [90/3072] via 192.125.30.1, 06:22:50, GigabitEthernet0/0
D       192.125.20.0/24 [90/3072] via 192.125.30.1, 06:22:50, GigabitEthernet0/0
    192.125.30.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.125.30.0/24 is directly connected, GigabitEthernet0/0
L       192.125.30.2/32 is directly connected, GigabitEthernet0/0
```

SWITCH_1

Do sh etherchannel summary

```
Switch(config)#do sh etherchannel summary
Flags: D - down          P - in port-channel
       I - stand-alone s - suspended
       H - Hot-standby (LACP only)
       R - Layer3        S - Layer2
       U - in use        f - failed to allocate aggregator
       u - unsuitable for bundling
       w - waiting to be aggregated
       d - default port
```

```
Number of channel-groups in use: 1
Number of aggregators:           1
```

Group	Port-channel	Protocol	Ports
1	Pol(SU)	LACP	Fa0/3(P) Fa0/4(P)

SWITCH_2

Do sh etherchannel summary

```
Switch(config)#do sh etherchannel summary
Flags:  D - down          P - in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port
```

```
Number of channel-groups in use: 1
```

```
Number of aggregators: 1
```

Group	Port-channel	Protocol	Ports
1	Pol(SU)	LACP	Fa0/3(P) Fa0/4(P)